Capstone Project Documentation

1. Introduction

This project focuses on building a complete end-to-end data analytics and business intelligence solution using Python and Power BI. The objective was to collect product-related data, clean and preprocess it, perform exploratory analysis, build analytical models, and finally create an interactive dashboard for business decision-making. The project integrates web scraping, data preprocessing, clustering analysis, AI-powered insights, and advanced Power BI features.

2. Problem Statement

Businesses dealing with product catalogs often struggle to understand customer preferences, pricing behavior, stock pressure, and product performance. The goal of this project was to analyze product data and generate actionable insights related to pricing, ratings, stock management, and product segmentation.

3. Web Scraping & Data Collection

Data was collected using Python-based web scraping techniques. A reproducible scraping script/notebook was developed to extract product-level information from the selected website/API.

- Python libraries such as BeautifulSoup, Requests, Pandas, and Selenium were used.
- Product attributes including Category, Price, Rating, and Stock Count were extracted.
- Data was stored in CSV format for further preprocessing.
- The scraping pipeline was designed to be reusable and scalable.

4. Data Cleaning & Preprocessing

The raw scraped dataset contained inconsistencies, missing values, duplicate records, and formatting issues. Data preprocessing was performed using Python and Power Query in Power BI.

- Removed duplicate records from the dataset.
- Handled missing and inconsistent values.
- Standardized column formats and data types.
- Removed invalid category entries such as 'Add a comment'.
- Created derived columns including Price Band, Rating Level, Stock Pressure, and Cluster Labels.
- Converted numerical fields into analysis-friendly formats.
- Validated data quality before dashboard integration.

5. Exploratory Data Analysis (EDA)

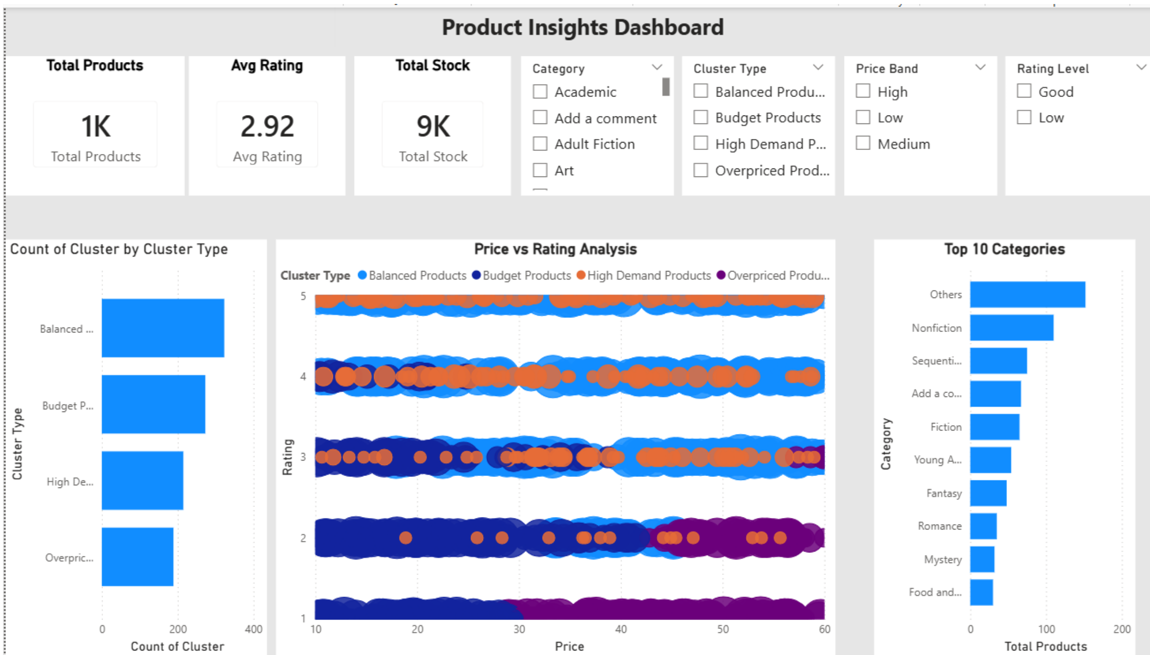
Exploratory Data Analysis was performed to identify hidden patterns, customer behavior trends, price distributions, and relationships between ratings and stock availability.

- Price distribution analysis across product categories.
- Rating analysis for different product clusters.
- Stock distribution by category and cluster type.
- Scatter plot analysis between Price and Rating.
- Detection of high-demand and overpriced product segments.
- Comparative analysis of Balanced, Budget, High Demand, and Overpriced product clusters.

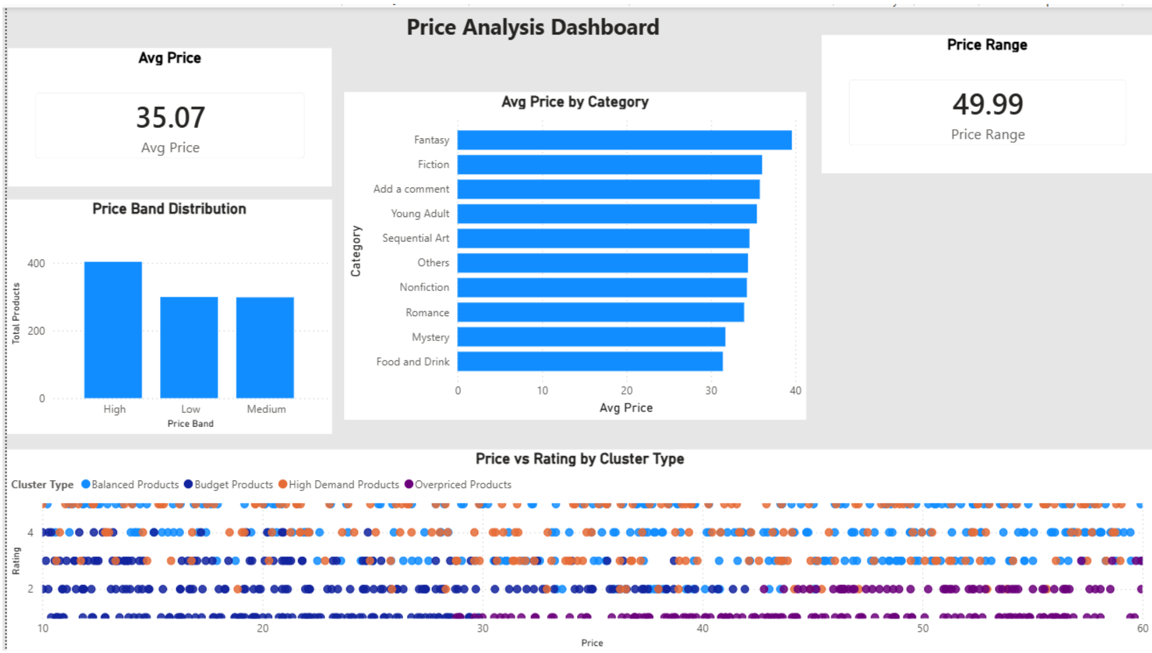
6. Data Visualization & Dashboard Design

Interactive dashboards were developed in Power BI to present insights in a visually appealing and business-friendly format.

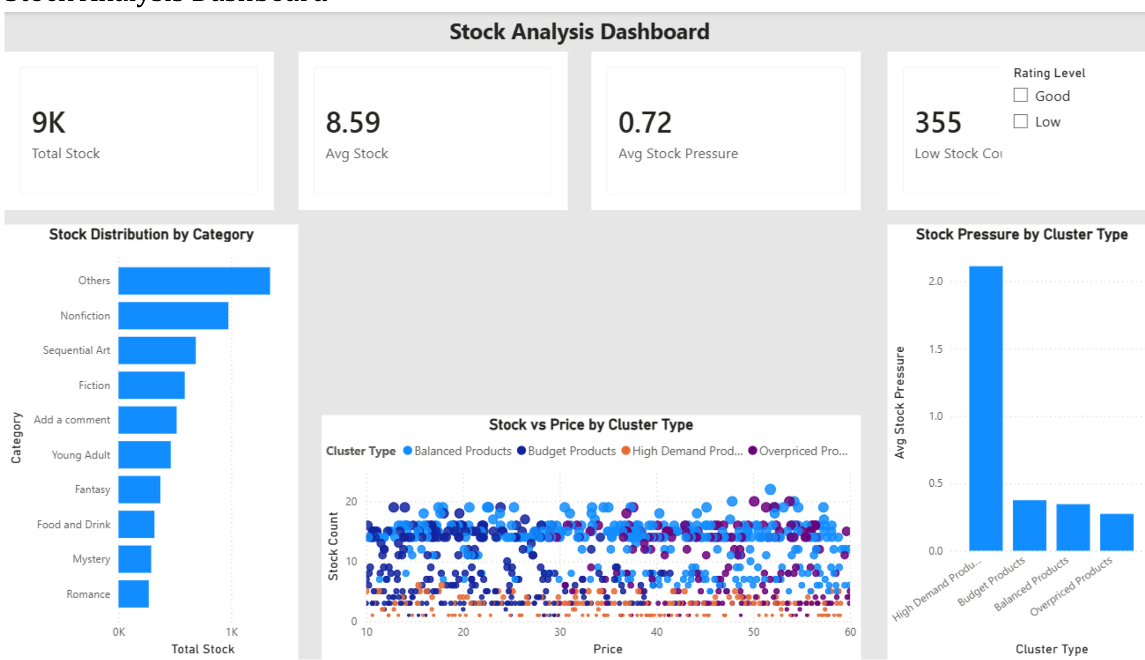
1. Main Dashboard



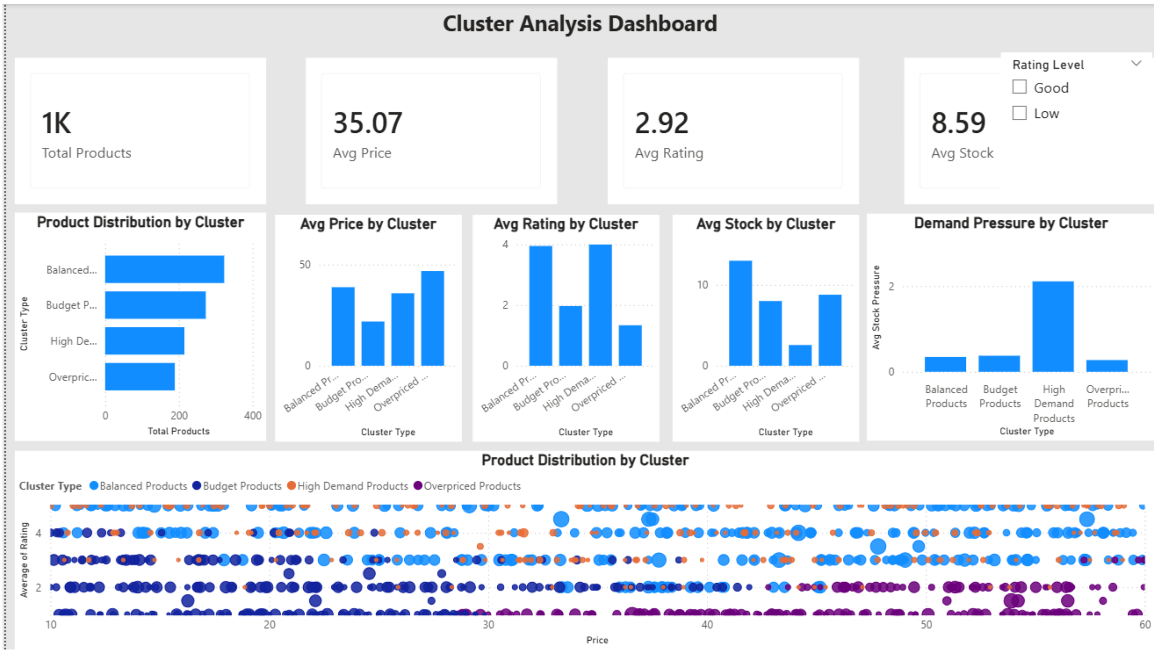
2. Price Analysis Dashboard



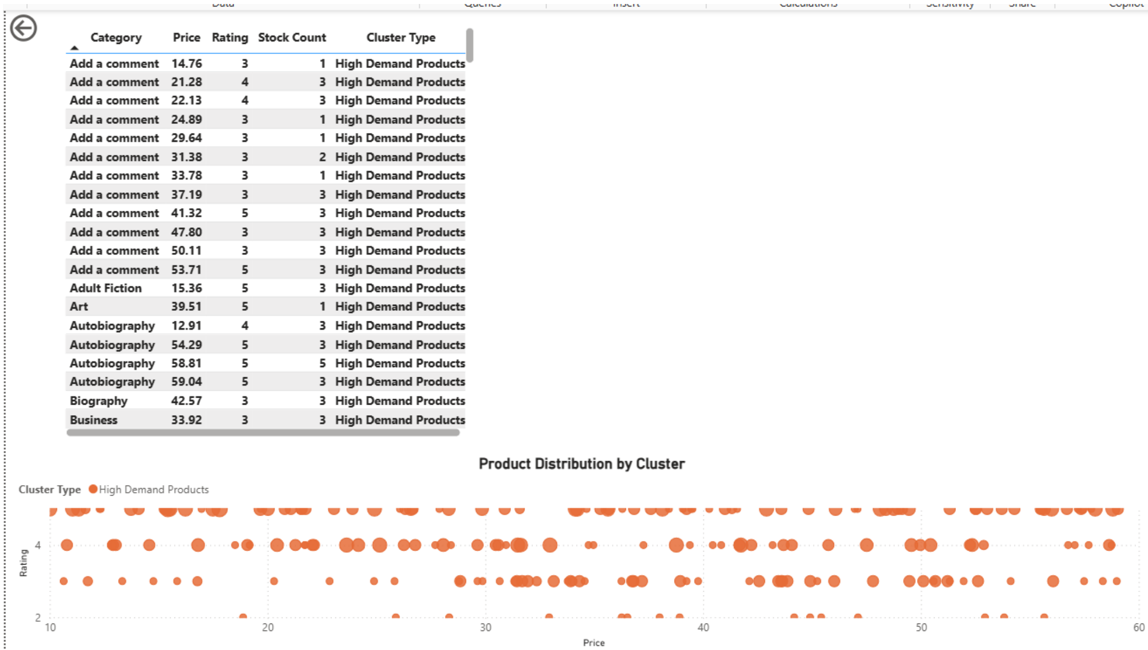
3. Stock Analysis Dashboard



4. Cluster Analysis Dashboard

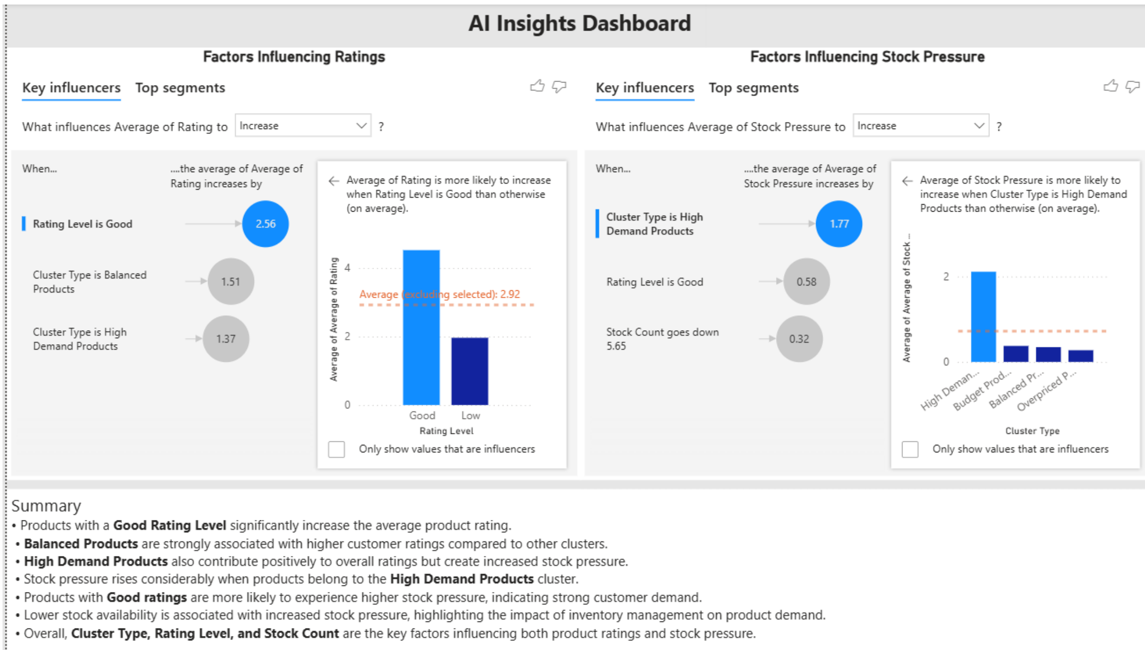


5. Cluster Drill-through Detail Page



Category	Price	Rating	Stock Count	Cluster Type
Add a comment	14.76	3	1	High Demand Products
Add a comment	21.28	4	3	High Demand Products
Add a comment	22.13	4	3	High Demand Products
Add a comment	24.89	3	1	High Demand Products
Add a comment	29.64	3	1	High Demand Products
Add a comment	31.38	3	2	High Demand Products
Add a comment	33.78	3	1	High Demand Products
Add a comment	37.19	3	3	High Demand Products
Add a comment	41.32	5	3	High Demand Products
Add a comment	47.80	3	3	High Demand Products
Add a comment	50.11	3	3	High Demand Products
Add a comment	53.71	5	3	High Demand Products
Adult Fiction	15.36	5	3	High Demand Products
Art	39.51	5	1	High Demand Products
Autobiography	12.91	4	3	High Demand Products
Autobiography	54.29	5	3	High Demand Products
Autobiography	58.81	5	5	High Demand Products
Autobiography	59.04	5	3	High Demand Products
Biography	42.57	3	3	High Demand Products
Business	33.92	3	3	High Demand Products

6. AI Insights Dashboard



7. Key Performance Indicators (KPIs)

- Total Products
- Average Price
- Average Rating
- Total Stock
- Average Stock Pressure
- Low Stock Count

8. Trend Charts & Visualizations Used

- Bar Charts
- Cluster Distribution Charts
- Scatter Plots
- Price Band Distribution Charts
- Stock Pressure Analysis Charts
- Category-wise Average Price Analysis
- AI Key Influencers Visual
- Drill-through Detail Tables

9. Model Building & Evaluation

An unsupervised machine learning clustering approach was used to segment products into meaningful groups based on pricing, rating behavior, and stock characteristics.

- Clustering technique was applied to identify product behavior patterns.
- Products were segmented into Balanced Products, Budget Products, High Demand Products, and Overpriced Products.

- Cluster analysis enabled business segmentation and strategic decision-making.
- Model results were integrated directly into the Power BI dashboard.

10. AI Visuals & Insights

Power BI AI visuals were implemented to automatically identify influential factors affecting product ratings and stock pressure.

- Key Influencers visual was used to identify factors influencing ratings.
- Balanced Products were associated with higher ratings.
- High Demand Products contributed significantly to stock pressure.
- Good Rating Levels increased average ratings and stock pressure.
- AI-generated insights improved business storytelling.

11. Advanced Power BI Features Implemented

- Sync Slicers
- Drill-through Analysis
- Role-Level Security (RLS)
- Interactive Filters
- AI Key Influencers Visual

12. Key Business Insights

- Balanced Products consistently achieved higher customer ratings.
- Overpriced Products showed comparatively lower ratings despite higher pricing.
- High Demand Products generated maximum stock pressure.
- Lower-priced products often maintained better customer satisfaction.
- Category and Cluster Type strongly influenced customer ratings and stock behavior.
- Fantasy, Fiction, and Young Adult categories showed higher average prices.
- Budget Products maintained moderate ratings with lower pricing.

13. Business Recommendations

- Optimize pricing strategies for Overpriced Products.
- Improve stock management for High Demand Products.
- Increase promotion of Balanced Products due to higher customer satisfaction.
- Use AI-driven insights for demand forecasting and inventory planning.
- Monitor low-stock products proactively to reduce stock-out risk.
- Leverage customer rating behavior to improve pricing and marketing strategies.

14. Challenges Faced

- Handling inconsistent scraped data formats.
- Managing category-level data cleaning.
- Designing effective clustering-based segmentation.
- Optimizing dashboard layout and visualization balance.
- Implementing advanced Power BI features such as RLS and drill-through.

15. Conclusion

This project successfully demonstrated the complete lifecycle of a data analytics solution, starting from web scraping and preprocessing to machine learning-based clustering and interactive dashboard development. The final Power BI solution provided actionable business insights using AI visuals, interactive reports, and advanced analytical features. The project highlights practical implementation of data analytics, business intelligence, and visualization techniques for real-world decision-making.

16. Web Scraping Code

```
import requests

from bs4 import BeautifulSoup

import pandas as pd


base_url = "http://books.toscrape.com/catalogue/page-{}.html"

main_url = "http://books.toscrape.com/catalogue/"


data = []


for page in range(1, 51):

  url = base_url.format(page)

  response = requests.get(url)

  soup = BeautifulSoup(response.text, "html.parser")


  books = soup.find_all("article", class_="product_pod")


  for book in books:

    title = book.h3.a["title"]

    price = book.find("p", class_="price_color").text

    rating = book.p["class"][1]
```

```
    # Get product link
    relative_link = book.h3.a["href"]
    product_link = main_url + relative_link

    # Visit product page
    product_response = requests.get(product_link)
    product_soup = BeautifulSoup(product_response.text, "html.parser")

    # Extract additional details
    try:
        description = product_soup.find("meta", attrs={"name":
"description"})["content"].strip()
    except:
        description = "No description"

    try:
        category = product_soup.find("ul", class_="breadcrumb").find_all("li")[2].text.strip()
    except:
        category = "Unknown"

    table = product_soup.find("table", class_="table table-striped")
    rows = table.find_all("tr")

    product_info = {}
    for row in rows:
```

```
            key = row.th.text
            value = row.td.text
            product_info[key] = value

        data.append({
            "title": title,
            "price": price,
            "rating": rating,
            "availability": product_info.get("Availability"),
            "UPC": product_info.get("UPC"),
            "product_type": product_info.get("Product Type"),
            "tax": product_info.get("Tax"),
            "num_reviews": product_info.get("Number of reviews"),
            "category": category,
            "description": description
        })
df = pd.DataFrame(data)
df.to_csv("cleaned_data.csv", index=False)
print("Done!")
```

17. Future Scope

- Integrate real-time API-based product data collection.
- Deploy dashboards using Power BI Service for enterprise sharing.
- Use predictive forecasting models for stock demand prediction.
- Implement recommendation systems using collaborative filtering.
- Enhance AI storytelling using advanced Copilot integration.

Video summary drive link :-
https://drive.google.com/drive/u/0/folders/1s70AZhGcWHKNMKJVvgYInirOVnuJ2MLd